

last class:

ordered pairs: (x, y) [two things separated by a comma & $()$]

Also, we know (ordering matters)

$$(2, 4) \neq (4, 2) \quad \boxed{\text{unlike set } \{2, 4\} = \{4, 2\}}$$

some examples of ordered pairs:

$$(i) (2, 3) \quad (ii) (H, T) \quad (iii) (\{2, 5\}, \{7\}) \quad (iv) (R, (0, 0))$$

$$(v) (2, \{1, 2, 3\})$$

Now, let's define Cartesian product:

Defⁿ: The Cartesian product of two sets $A \in B$ is a set of ordered pairs, defined as,

$$A \times B = \{ (a, b) : a \in A, b \in B \}$$

$$\text{example: } A = \{k, l, m\} \quad B = \{p, q\}$$

$$A \times B = \{ (k, p), (k, q), (l, p), (l, q), (m, p), (m, q) \}$$

A		<u>k</u>	<u>l</u>	<u>m</u>
B	<u>p</u>	(k, p)	(l, p)	(m, p)
	<u>q</u>	(k, q)	(l, q)	(m, q)

Also, let, $A = \{ \begin{smallmatrix} \square \\ 1 \end{smallmatrix}, \begin{smallmatrix} \square \\ 2 \end{smallmatrix}, \begin{smallmatrix} \square \\ 3 \end{smallmatrix}, \begin{smallmatrix} \square \\ 4 \end{smallmatrix}, \begin{smallmatrix} \square \\ 5 \end{smallmatrix}, \begin{smallmatrix} \square \\ 6 \end{smallmatrix} \}$

So, $A \times A$

A	1	2	3	4	5	6
1	(1,1)	(2,1)				
2	(1,2)	(2,2)				
3	(1,3)	(2,3)				
4	(1,4)	(2,4)				
5	(1,5)	(2,5)				
6	(1,6)	(2,6)				

$$6 \times 6 = 36$$

The two examples should motivate you with
Cardinality of a Cartesian Product:

$$|A \times B| = |A| \cdot |B|$$

(This theorem is for finite sets)

Problem:

$$\textcircled{1} A = \{\pi, e, o\} \text{ \& } B = \{0, 1\}$$

$$(i) A \times B$$

$$(ii) \emptyset \times B = \emptyset$$

$$(iii) \{\emptyset\} \times B$$

$$\{\emptyset\} \times B = \{(a, b) \mid a \in \{\emptyset\}, b \in B\}$$

$$[A \times B = \{(a, b) \mid a \in A, b \in B\}]$$

$$\emptyset \times B = \{(a, b) \mid a \in \emptyset, b \in B\}$$

$$\textcircled{2} \{x \in \mathbb{R} : \underline{x} = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$$

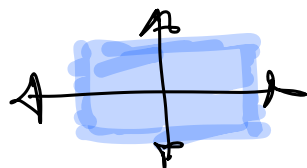
$$= \{ -\sqrt{2}, \sqrt{2} \} \times \{ -2, 2 \}$$

$$\textcircled{3} \{x \in \mathbb{Z} : x = 2\} \times \{x \in \mathbb{R} : |x| = 2\}$$

$$= \emptyset \times \{ -2, 2 \} = \emptyset$$

shifting gears to infinite set:

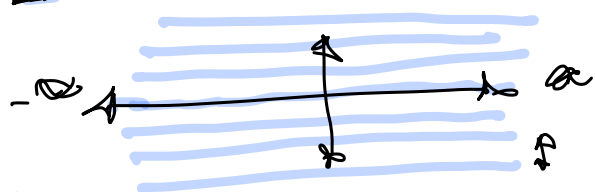
$$(i) \mathbb{R} \times \mathbb{R} = \{ (x, y) : x \in \mathbb{R}, y \in \mathbb{R} \}$$



$$(ii) \mathbb{R} \times \mathbb{Z}$$

$$(iii) \mathbb{Z} \times \mathbb{R}$$

$$\bullet \mathbb{R} \times \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$$



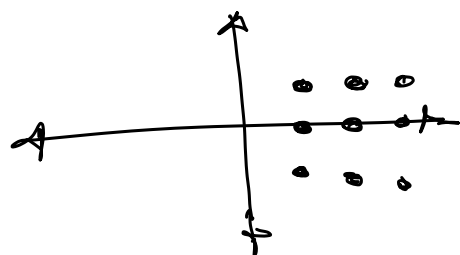
Problems:

$$\bullet \text{ sketch: } \{1, 2, 3\} \times \{-1, 0, 1\} \text{ in } \mathbb{R}^2$$

$$\bullet \text{ sketch: } \{0\} \times \mathbb{N} \text{ in } \mathbb{R}^2$$

$$(i) \{ (1, -1), (1, 0), (1, 1), (2, -1), (2, 0), (2, 1),$$

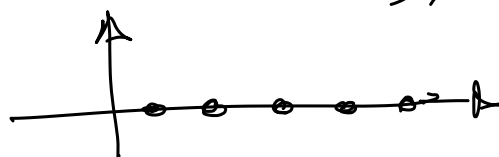
$\dots \}$



$$(ii) \{0\} \times \mathbb{N} = (0, \mathbb{N})$$

$$\text{in } \mathbb{R}^2, \mathbb{N} = \{1, 2, 3, 4, \dots\}$$

$$\{(0, 1), (0, 2), (0, 3), \dots\}$$



Cartesian product is not restricted to two sets

we can take Cartesian product of 3 sets, (ordered triple)

say: $A \times B \times C$, $A \times B \times C \times D$, $A \times B \times C \times \dots \times$
Ordered triple Ordered quadruple Ordered n-tuple

short note:

$$\mathbb{R} \times (\mathbb{N} \times \mathbb{Z}) \neq \mathbb{R} \times \mathbb{N} \times \mathbb{Z}$$

$(\sqrt{3}, (4, -7))$ $(\sqrt{2}, 2, -2)$

We can extend beyond 3,

$$\tilde{A}_1 \times \tilde{A}_2 \times \dots \times \tilde{A}_n = \{ (x_1, x_2, x_3, \dots, x_n) : x_1 \in A_1, x_2 \in A_2, x_3 \in A_3, \dots \}$$

where A_i are sets,

for any set A and n a positive integer

$$A^n = \underbrace{A \times A \times \dots \times A}_n = \{ (x_1, x_2, x_3, \dots, x_n) : x_1, x_2, x_3, \dots, x_n \in A \}$$

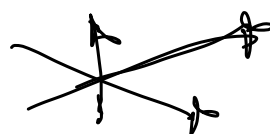
(ordered n-tuple)

examples:

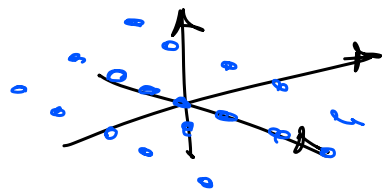
• $\mathbb{R}^2 : \{ (x, y) : x, y \in \mathbb{R} \}$



• $\mathbb{R}^3 : \{ (x, y, z) : x, y, z \in \mathbb{R} \}$



(Cartesian power)



Problems:

(1) $B = \{0, 1\}$ find, $B \times (B \times B)$ and $\underline{B \times B \times B} = B^3$

(ii) $\{2\phi_3 \times 2\phi_3 \times 2\phi_3\}$

(iii) sketch of $[0,1] \times \mathbb{R}^2 \times [1,2]$

(i) $B = \{0, 1\}$, $B \times B = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

$$\begin{aligned} \mathbb{A}^1 \times (\mathbb{A}^1 \times \mathbb{A}^1) &= \{0, 1\} \times \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ &= \{(0, (0, 0)), (0, (0, 1)), (0, (1, 0)), \\ &\quad \dots \dots \dots \} \end{aligned}$$

$$\rightarrow B \times B \times B = \{ (x, y, z) : x, y, \underline{z} \in B \}$$

$$= \{ (0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1) \}$$

$$\begin{bmatrix} B = 20, 12 \\ \hat{z} = \overset{x}{0}, \overset{x}{1} \\ y = \overset{\check{0}}{0}, \overset{\check{1}}{1} \\ x = \overset{\check{0}}{0}, \underline{\underline{1}} \end{bmatrix}$$

subsets:

lets start with a set,

$$B = \{0, 1, 2, 3, 4\}$$

we can construct a set by taking a subcollection from the above set, $A = \{0, 1, 2\} \rightarrow$ this is also a set

Defⁿ: suppose A and B are sets. If every element of A is also an element of B , then we say:

$$A \subseteq B$$

else,

$$A \not\subseteq B$$

In that case, A is not the subset of B

B does not contain all the elements of A

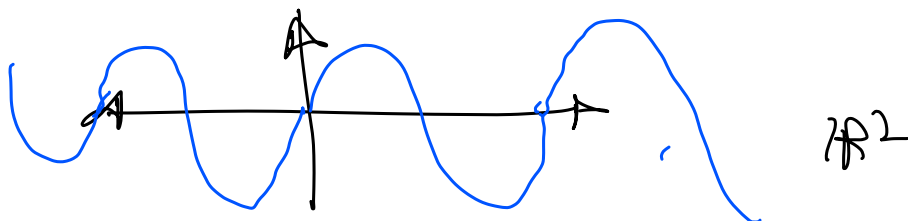
examples:

$$(i) \{2, 3, 7\} \subseteq \{1, 2, 3, 4, 5, 6, 7\}$$

$$(ii) \{2, 3, 7\} \subseteq \{2, 3, 7\}$$

$$(iii) \{(x, \sin x) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$$

graph:



$$(iv) \{3, 5, 7, 11, 13, 17, \dots\} \subseteq \mathbb{N}$$

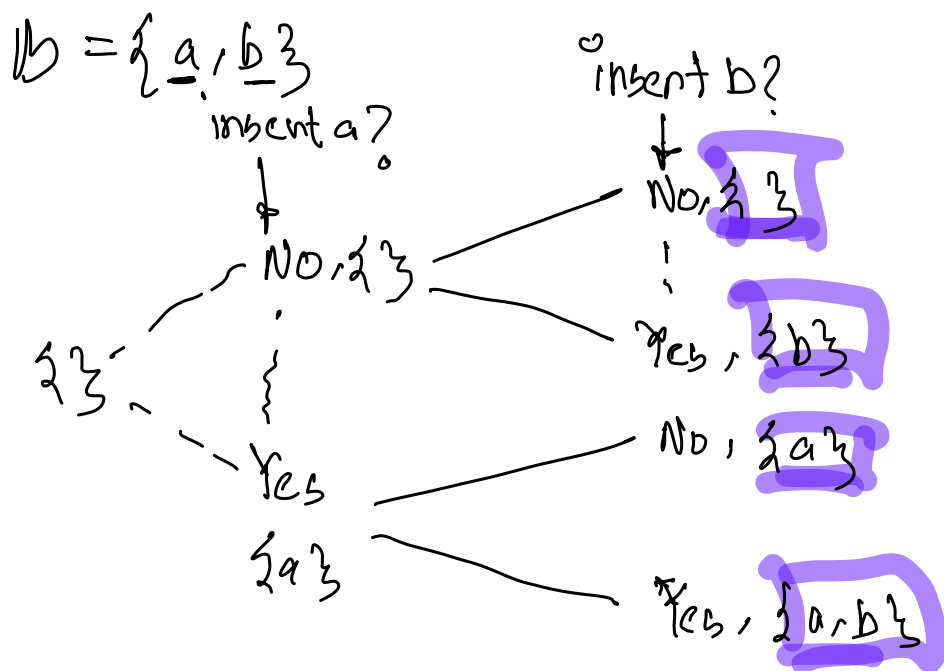
$$(v) \{1, 3, 5, 7, \dots\} \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

$$(vi) A \subseteq A \text{ [for any set } A]$$

$$(vii) \emptyset \subseteq A \text{ [for any set } A], \text{ why?}$$

every element of $\emptyset$ is also an element of A
(vacuously true)

Construction of subset of a set:
using the tree diagram



$$4 = 2^2$$

no of subsets

in general: for a A with, $|A| = n$, there are 2^n no of subsets,

$$\text{for, } B = \{1, 2, \{1, 3\}\}$$

Caution:

$$\{1, 3\} \notin B, \{1, 3\} \in B$$

$$\{1\} \notin B, \{1\} \subseteq B$$

some more Caution:

(i) in $\{1, \{1\}\}$

- $\{1\} \in \{1, \{1\}\}$
- also, $\{1\} \subseteq \{1, \{1\}\}$
- and, $\{\{1\}\} \subseteq \{1, \{1\}\}$

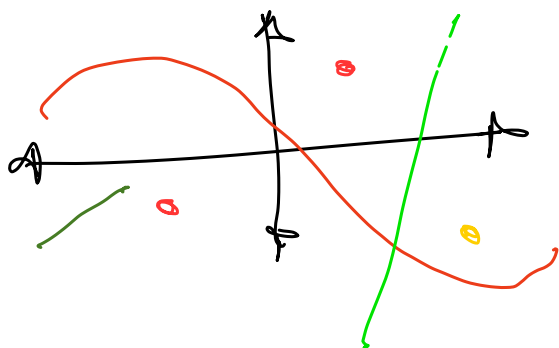
(ii) in $\{\emptyset, \mathbb{N}\}$

- $\mathbb{N} \notin \{\emptyset, \mathbb{N}\}$, $\mathbb{N} \in \{\emptyset, \mathbb{N}\}$
- $\{\mathbb{N}\} \subseteq \{\emptyset, \mathbb{N}\}$

(iii) in $\mathbb{N} \times \mathbb{N}$, = $\{(x, y) : x \in \mathbb{N}, y \in \mathbb{N}\}$

- $(2, 3) \in \mathbb{N} \times \mathbb{N}$
- $\{(2, 3)\} \subseteq \mathbb{N} \times \mathbb{N}$

subsets of $\mathbb{R}^r$:



$$\mathbb{R}^r = \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$$

- collection of points
curves etc on $\mathbb{R}^r$
are subsets of $\mathbb{R}^r$

exercise:

- Determine true or false:

$$\{(x, y) \in \mathbb{R}^r : \underline{x^r - x = 0}\} \subseteq \{(x, y) \in \mathbb{R}^r : x-1=0\}?$$

$$\{(0, \mathbb{R}), (1, \mathbb{R})\}^{x=0,1} \supseteq \{(1, \mathbb{R})\}$$

- $\mathbb{R}^n \leq \mathbb{R}^3$? (think about it)

Power sets:

Defⁿ: If A is a set, then the power set of A is a set which consists of, all the subsets of A

$$\mathcal{P}(A) = \{ \text{All possible subsets of } A \}$$

For a finite set A , there are $2^{|A|}$ no of subsets.

so, $|\mathcal{P}(A)| = 2^{|A|}$

example: $\mathcal{P}(\{ \{a, b\}, \{c\} \}) = \{ \emptyset, \{a, b\}, \{c\}, \{ \{a, b\}, \{c\} \} \}$

$\underbrace{\qquad\qquad\qquad}_{2^2=4}$

• Also, $\mathcal{P}(\{ \emptyset, 5 \}) = \{ \emptyset, \{ \emptyset \}, \{ 5 \}, \{ \emptyset, 5 \} \}$

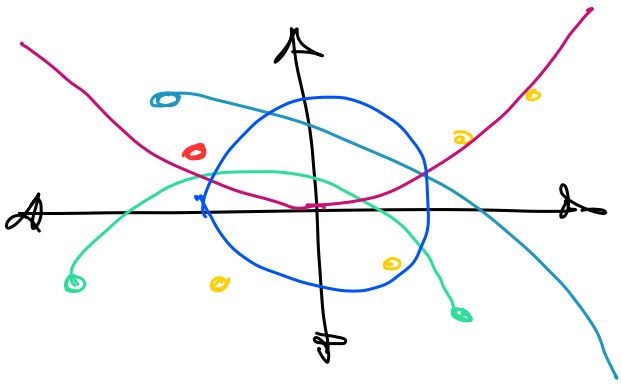
• compute: $\mathcal{P}(\{1, 2\}) \times \mathcal{P}(\{3\})$

$$= \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \} \times \{ \emptyset, \{3\} \} = \{ (\emptyset, \emptyset), (\{1\}, \emptyset), (\{2\}, \emptyset), \dots \}$$

• compute $\mathcal{P}(\{\mathbb{Z}, \mathbb{N}\})$

$$= \{\emptyset, \{\mathbb{Z}\}, \{\mathbb{N}\}, \{\mathbb{Z}, \mathbb{N}\}\}$$

Power set of $\mathbb{R}^2$:



$\mathcal{P}(\mathbb{R}^2) = \{ \text{collection of all possible subsets of } \mathbb{R}^2 \}$

$$\{x : x \in \mathcal{P}(A), |x| = 1\}$$

• for $|A|=3$, $|B|=4$

(i) Find $|\mathcal{P}(A) \times \mathcal{P}(B)| = ?$ (ii) Find $|\mathcal{P}(A \times \mathcal{P}(B))|$

(iii) $|\mathcal{P}(\mathcal{P}(A \times \emptyset))|$ (iv) $|\{x \subseteq \mathcal{P}(A) : |x| \leq 1\}|$

$$(i) \mathcal{P}(A) = 2^{|A|} \quad \mathcal{P}(B) = 2^{|B|}$$

$$|\mathcal{P}(A) \times \mathcal{P}(B)| = |\mathcal{P}(A)| \cdot |\mathcal{P}(B)| = 2^{|A|} \cdot 2^{|B|}$$

$$(ii) A \times P(B) = \underline{3 \cdot 2^4}, \quad \underline{P(A \times P(B))} = 2^{3 \cdot 2^4}$$

$$(iv) A = \{a, b, c\}, \quad \underline{P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \dots\}}$$

$$= 4$$

Union, Intersection, Difference

we start with two sets A & B ,

union: A

intersection:

difference:

Problems:

$$A = \{a, b, c, d, e\}, \quad B = \{d, e, f\}, \quad C = \{1, 2, 3\}$$

$$\bullet A - B =$$

$$\bullet B - A =$$

note

$$[A \cup B = B \cup A, \\ A \cap B = B \cap A]$$

$$\bullet (A - B) \cup (B - A) =$$

$$\bullet (A \cap B) \times B =$$

some more problems:

$$A = \{0, 1\}$$

$$B = \{1, 2\}$$

- $\mathcal{P}(A \times B)$

- $\mathcal{P}(A) \cap \mathcal{P}(B)$

intervals:

- $([2, 5] \cup [1, 6]) - [8, 10]$

- $[2, 3) \cup (3, 5]$

- $[2, 3] \cap (3, 5]$

- $[0, 3] - [1, 2]$

• example: $A = \{(x, x^2) : x \in \mathbb{R}\}$

Also, $B = \{(x, y) : y = x+1, x \in \mathbb{R}\}$

(i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$

• Think about:

example: $A = \{(x, x^2) : x \in \mathbb{N}\}$

Also, $B = \{(x, y) : y = x+1, x \in \mathbb{R}\}$

(i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$

• sketch, $X = [-1, 3] \times [0, 2]$ $Y = \{1\} \times [1, 2]$

Now locate, $X \cap Y$, $X - Y$

Compliment:

Universal set:

When dealing with sets, in almost all the cases, that set is a part of a larger set.

Say, the set of Primes, P

$$P =$$

Now, for, Circle in $\mathbb{R}^n$

Defn of compliment:

Let A be a set, with U universal set.

The compliment of A ,

$$\bar{A} =$$

Problems:

$$A = \{0, 2, 4, 6, 8\}, B = \{1, 3, 5, 7\}$$

$$U = \{0, 1, 2, 3, \dots, 8\}$$

(i) $A \cap \bar{A}$

(ii) $\overline{A \cup B}$

(iii) $\bar{A} \cap \bar{B}$

(iv) $A - \bar{A}$

(v) $\bar{A} \times B$

(vi) $\overline{A \cap B}$

(vii) $\bar{A} \cup \bar{B}$

• Sketch: for $X = \{1\} \times [2, 4]$

(i) Find, $\bar{X}$

(ii) Also, Find: $\bar{X} \cap ([0, 2] \times [0, 3])$